SkillForge: Self-Distilling Agents for Project-Specific Issue Resolution

Silin Chen*, Han Li*, Xiaodong Gu†, Yuling Shi, Haibing Guan · Shanghai Jiao Tong University · arXiv 2608.18933v1 · ‡ = computed by us

HarnessRSI 69 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

1 Synthesize issues
passing tests → coverage trace → LLM rewrites the critical segments with the original masked → tests fail = buggy instance + problem text

2 Agent attempts them
Mini-SWE-Agent solves each synthetic issue; successful and failed trajectories are kept

3 Distill entity-linked skills
LLM turns trajectories into skills tied to repo files / functions (AST index)

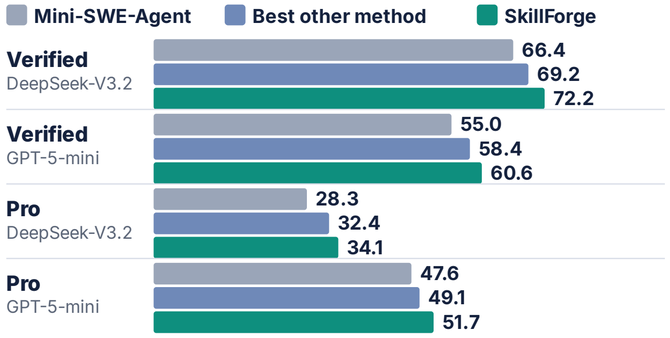
M_ext · global	M_int · local
purpose · playbook · related APIs	**how to edit the entity · pitfalls (from failed runs)**
4 · Use at inference: top-5 by BM25 on the issue text, put at the start of the prompt	4 · Use at inference: injected as an extra observation when the agent touches the file

weights never change; kept artifact = skill store
from code + tests only; no iteration of the pipeline described

Reading notes

- Verified: synthesis is redone per target instance on a rolled-back snapshot; only earlier-commit skills.
- Entity alignment also adds files from "the golden patch"; the paper does not say whether it is the synthetic one.

2 Headline: +4.1 to +5.8 over the plain agent



Pass@1 (%). Verified: 500 tasks; Table I states a 3-run mean, yet values look like whole task counts; no SD. Pro: 731 tasks, 2 baselines. Best other on GPT-5-mini Verified is EvoCoder (text says MemGovern, 58.0).

4 Where the gain is absent or thin

- **Margin over the best other method: 1.7–3.0 points** (Verified +3.0, 2.2‡; Pro +1.7, +2.6); one Verified task = 0.2‡; no test vs the best baseline
- **Other LLM's skills: no gain** (65.2 < 66.4; 55.0 = baseline); 1 benchmark, 2 models
- Fig. 5: **0.0%** on DeepSeek Astropy; SWE-Exp ties or beats SkillForge in some cells; no per-repo counts
- k sweep on evaluated repos (data look like Django alone)
- **Not reported:** equal-spend baseline; SD; test behind †

3 Controls that decide if the gain is real

Ablations and same-injection variants (Tables I, III)

Verified Pass@1 (%)	DeepSeek	GPT-5-mini
Mini-SWE-Agent	66.4	55.0
w/o global skills (Tab. III)	68.4	57.6
w/o local skills (Tab. III)	67.8	57.2
Same injection, SWE-Smith rewrites	68.0	56.4
Same injection, LLM repo summary	68.7	**54.4**
SkillForge (full)	**72.2**	**60.6**

All below full; all above the plain agent except LLM-summary on GPT-5-mini (54.4 < 55.0). Runs per cell not stated.

Cross-LLM skills (Table IV)

Resolver · skills from	own	other LLM	no skills
DeepSeek · GPT-5-mini	**72.2**	**65.2**	66.4
GPT-5-mini · DeepSeek	**60.6**	**55.0**	55.0

Cost, offline stage included

Avg cost / issue (Verified)	DeepSeek	GPT-5-mini
Mini-SWE-Agent	$0.049	$0.031
SkillForge (offline stage amortized)	$0.074	$0.066

Dearer than Mini-SWE-Agent; no equal-spend row for baselines.

Held-out (Verified): per target, repo rolled back before the gold patch; only earlier-commit skills. Pro: protocol not described.

Takeaway

Skills distilled from bugs made out of a repo's own tests lift a plain agent 4–6 points, and both skill sets matter. The edge over other methods is 1.7–3.0 points, with no spread reported; in the one cross-model test, the other LLM's skills gave no gain.

Our read · HarnessRSI

Score 69 · ★★ · evidence level: retained

Reports: time-isolated eval, ablation, 3 runs, cost, cross-LLM.
Does not report: matched-compute baseline, SD, cross-repo transfer.